

**PROPOSING THE BEST MACHINE LEARNING MODEL FOR
MAPPING THE STOCK MARKET DYNAMICS: LISTED
COMPANIES DILEMMA**



By

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A thesis presented to Bahria University, Islamabad in partial fulfillment of the requirements
for the degree of Master in Business Analytics

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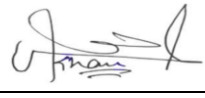
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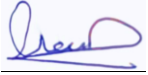
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ABSTRACT

This study utilises ML methods to predict the behaviour of stock markets using share price dynamics with reference to the behaviour of the S&P 500 and NASDAQ indices in a ten-year timeline (2015 years to 2024). Filling in an existing literature gap on emerging markets in the literature, the study performs a comparative study of eight supervised machine learning algorithm, including Support Vector Regression (SVR), Artificial Neural Network (ANN), k-Nearest Neighbors (KNN), Naive Bayes, Random Forest (RF), Recurrent Neural Network (RNN), Long Short-Term Memory (LSTM) and a Two-Stage Hybrid SVR-RF model through a common methodological approach. Historical price data (Open, High, Low, Close) was obtained daily and pre-processed, by employing Min Max normalization, as well as by augmenting with technical indicators. Models were coded in Python, hyperparameter tuned and the results assessed by numerous metrics: Root Mean Square Error (RMSE), Mean Absolute Error (MAE), Mean Bias Error (MBE), Mean Absolute Percentage Error (MAPE), R^2 , and Directional Accuracy. Results indicated that the Two-Stage Hybrid SVR-RF model is well able to obtain the lowest RMSE and the closest-to zero bias as compared to the other three models with LSTM next in both indices due to its capability to learn sequential dependencies. ANN and RF gave moderate performance and SVR, as well as Naive Bayes, were not so satisfactory. The results support the significance of a hybrid ML method in enhancing the quality of forecasting in unstable financial markets. The research is theoretically useful, as the study elucidates the existence of differences in the performance of models, and practically useful, as it can provide evidence on aspects of guidance to the investors, analysts and policymakers who wish to have a robust market prediction tool.

Keywords: stock market forecasting, machine learning, stock price dynamics, random forest, long short-term memory, financial prediction

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1. INTRODUCTION

1.1 BACKGROUND

Over the past decade, financial markets have become significantly more intricate shaped by the rapid expansion of data availability and the increasing influence of real-time investor sentiment. While relying on stock market predictions, researchers find that various traditional models do not detect changes in volatility or prices when dramatic or sensational events occur (Choudhury & Garg, 2021). Predicting stock market movements has always been a significant challenge in the finance arena. Stock prices are driven by the interplay of multiple economic indicators, political events, and investor behaviour, making it exceedingly hard to predict their future trajectory (Jena & Majhi, 2023). Accurate prediction is highly desired because it allows investors and institutions to develop strategies that decrease risk and enhance returns (Lin & Lobo Marques, 2024).

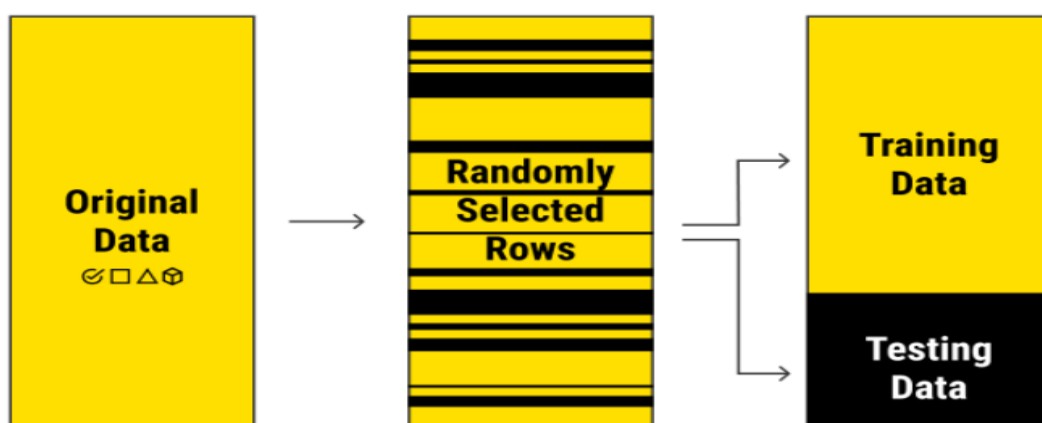
Recently, machine learning (ML) has gained momentum as a way of overcoming this prediction barrier. Many researchers are utilizing ML algorithms, with their ability to learn patterns from historical data, to approximate the subtle and non-linear characteristics of financial markets, characteristics that attritional models typically neglect. Stock indices like the S&P 500 and NASDAQ Composite are useful representatives of movement in markets since they group together the performance of hundreds (or even thousands) of listed companies. Instead of needing to analyse each firm, market trends can be derived by monitoring major stock indices. This chapter establishes the framework for the study by explaining the background related to the challenges of predicting stock markets, defining specific terms, and outlining the motivation, gaps, and aim of the research. More specifically, we introduce the "listed companies' dilemma," when faced with the uncertainty of the most appropriate ML model to map the dynamics of share price, and subsequently outline the research questions aimed at addressing this dilemma.

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1.2 DEFINITION OF CONCEPTS

1.2.1 Machine Learning

Machine Learning (ML) involves the development of algorithms that allow computer systems to identify patterns within data and make informed decisions or predictions, all without the need for explicit programming instructions. In the context of this study, ML techniques are applied to financial datasets to predict stock market movements by analysing historical share price trends. Unlike conventional statistical approaches, machine learning can handle extensive and complex data structures, uncover nonlinear relationships, and adjust to evolving patterns within the data (Khan et al., 2021).



Some of the most popular libraries for Python such as Scikit-learn, TensorFlow, XGBoost and Keras, assist in running different machine learning algorithms. When looking at time-series data, Random Forest, Support Vector Machines and LSTM from Recurrent Neural Networks (RNNs) are often used for predicting finances (Fischer & Krauss, 2018).

1.2.2 Stock Market Dynamics

Analysing stock market dynamics requires looking at shifts in stock prices and the influences of volatility, trends, the trading volume and changes in price returns. These factors change due to the company's economic results, the overall condition of the economy, investors' moods, noteworthy international events and important technology-based trends. In any market or industry, these factors play an important role in making financial forecasts. In this step, market

behavior is analysed by machines for the purpose of predicting changes in the stock market. Pakistan's KSE-100 Index and the S&P 500 Index in the US could be impacted by the performance of main listed corporations and changes in trending sectors. To create a reliable forecast or model, one must look at recent trends in share prices (Ahmed et al., 2019).

1.2.2 Listed Companies

Listed companies are companies that have made their shares available to the public through a renowned stock exchange and overseeing financial authorities. Regularly sharing their financial and operational results allows these companies' data to be more accessible to analysts. Listed companies are the linchpin of stock indices: indices, for example, the S&P 500, simply represent portfolios of listed company shares, while the level or performance of an index is derived from the prices of the constituents' listed companies. So, for example, a large-tech company that is listed on, say, the NASDAQ, will most certainly have a pronounced effect on the movement of the NASDAQ Composite. Movement of the stock market index is therefore a function of the collective action of the listed companies, which is brought about by their profit-making performance, growth prospects, the perceptions of the investors, etc., the dynamics of an index reflect these aspects. This is part of the reason and justification why stock indices are often referenced as a proxy for how "listed companies" are performing. An important consideration is that indices are weighted, so the movement of large companies will impact index movements more than small companies. Nevertheless, an index reflects the collective sentiment of the market toward the included market listed companies. The relevance of share price dynamics around the listed companies in the context of our study is that understanding them will assist in mapping and forecasting the movement of the stock market index itself.

1.3 RESEARCH MOTIVATION

The motivation for this research comes from the large potential benefits of improved accuracy in stock market forecasting and the constraints of existing prediction techniques. Financial markets are critical to investment and economic decision-making, and even slight

improvements in prediction accuracy can yield better investment behaviour, avoid risks, or re-allocate resources. For all investors, if there was a way to reliably forecast markets, there would be an opportunity for greater investment returns and reduced risk. Better quality forecasting has the potential to help develop strategies that “make better financial decisions, make profits, and reduce risk” for both individuals and institutions, (Lin & Lobo Marques, 2024). The need for better financial forecasting was exacerbated with the volatility experienced over the last few years which highlighted the disconnect between traditional forecasting models not providing warning for the sharp shifts. There is a strong incentive to construct models that could predict such movement and dynamics.

Examples of conventional forecasting methods, linear modelling time series or merely relying on fundamental analysis, have only a limited effectiveness in dealing with the complex nature of the stock market patterns and shifts. Often these are unable to quantify abrupt changes or deal with non-linear effects, leading to the exploration of models using machine learning, which could be used effectively on complex patterns. They thus, can produce predictive outcomes as decision rules could utilize information or patterns that simpler, analysable models by humans could miss. For instance, a deep learning model could identify some early warning signals with respect to how a group of tech stocks are holding in synchrony and are signalling a broader market shift, which a human could miss until after the fact. Initial work using ML to develop stock prediction systems suggest that predictive performance was increased with ML methods, and specifically ML models include sentiment analysis which enhanced predictive performance (Shah, Isah, & Zulkernine, 2019). These initial efforts lead to further exploratory work so if one ML model was better than a benchmark at one study setting, could a different ML model be better, or more stable for multiple study settings?

Consideration of the vast increase in data and computational capacity over the last few years is additional significant motivating factor. High frequency trading data, alternative datasets (e.g. social media sentiment, Google trends, etc.), powerful GPUs, could all now be considered possibilities for the prediction problem space. Given what we now financially consider about the synergies between finance and big-data analytic, we are not only able to tackle this problem with degree of feasibility with the techniques used in these locales, but we are almost at a point of interrogating the data as we consume it. We have a strong conviction, supported by previous

findings, that a ML approach will allow us projects that are more accurate and robust prediction of stock market dynamics than we previously imagined.

On a personal level, this research is driven by an interest in applying Python programming and its associated machine learning libraries to solve practical financial problems. Python's broad functionality, ease of use, and strong community support make it a vital tool for analysts working with large financial datasets. The goal of this study is to meet the need for good financial strategies by choosing the most suitable ML model to predict changes in stock prices.

1.4 GAP ANALYSIS

Over the past decade, a wide range of studies have explored the application of machine learning techniques in financial forecasting, particularly in developed economies such as the United States, the United Kingdom and China. These studies have shown that machine learning algorithms like Support Vector Machines, Random Forests and Long Short-Term Memory networks outperform traditional linear models in predicting stock prices and market trends (Fischer & Krauss, 2018; Patel et al., 2015). However, despite the growing body of literature in this domain, several critical gaps still remain particularly in the context of emerging markets like Pakistan.

First, there is limited research that directly compares the predictive accuracy of multiple ML models on the same dataset of stock market data based on share price dynamics. The research done so far usually examines one model or merely looks at macroeconomic data without focusing on daily closing prices, the trade volume or moving averages found in microeconomic data. It reduces the ability to judge which ML model is best for predicting share price movement with price information.

Many prior studies concentrate on a narrow set of markets or indices. For instance, a recent systematic review observed that most stock market prediction studies focus on just three countries, the USA, China, and Taiwan, with far fewer studies examining other markets (Kumbure et al., 2022). This skew means that findings might be market-specific; a model found to be best for U.S. stocks might not hold the top spot for European or emerging markets, simply

because it hasn't been tested widely. Moreover, even within the U.S., research often zeroes in on one index at a time. The lack of cross-market and multi-index studies represents a gap; our understanding of model performance is fragmented and might not generalize beyond the specific context of each study.

Additionally, not much effort has gone into objectively measuring models by using common performance indicators such as Root Mean Square Error, Mean Absolute Error, R^2 Score and classification accuracy for predictions about market direction. To find out which ML model is reliable enough, experts need to compare the measurements and metrics to choose the best for practical forecasting applications.

Many universities and research centers continue to analyze finance using Excel or SPSS, although Python is commonly used worldwide. This research seeks to bridge that gap by fully integrating Python and its machine learning libraries into the predictive modelling process providing both a technical and academic contribution to the field of Business Analytics.

1.5 PROBLEM STATEMENT

Stock markets are characterized by high volatility, unpredictable behavior and intricate structures. It is important for investors, policymakers and researchers to observe and understand trends in the market. Often, the activity of the market can be understood by looking at how a company's share prices increase and decrease each day. Even so, due to GARCH and ARIMA not being able to spot unusual trends or respond quickly to sudden market changes, they are currently less accurate than previously (Choudhury & Garg, 2021).

This research aims to check if changes in share prices can be used to forecast the stock market's performance with the help of machine learning. Experts found that ML works well in developed nations, yet few studies have compared its performance in Pakistan's stock market (Arif & Shah, 2022). Many local studies rely on traditional tools or apply a single ML model without evaluating performance across different algorithms using standardized metrics such as RMSE, MAE or R^2 . This creates a clear research gap. It is still unclear which ML model provides the most reliable predictions based on share price dynamics. The purpose of this research is to apply

various Python based ML models including Random Forest, Support Vector Regression and Long Short-Term Memory networks to historical stock data of listed firms. Using performance evaluation methods, the prediction quality of each model will be assessed and compared.

By addressing the questions of whether share price behavior can indicate broader market trends and which model offers the highest prediction accuracy, this research will help fill a gap in the application of machine learning to local stock markets. It will also provide a data-driven foundation for analysts, traders and researchers who aim to use ML for forecasting.

1.6 RESEARCH QUESTIONS

Can share price dynamic predict the overall market performance through Machine Learning techniques?

Which Machine Learning model offer the highest predictive accuracy based on performance evaluation models?

1.7 RESEARCH OBJECTIVES

The objectives of this research study are:

To evaluate the prediction qualities of Machine Learning techniques for the performance of Stock market based on share price dynamic.

To propose the best Machine Learning model for stock market prediction based on share price dynamics by using different performance evaluation models.

1.8 APPLICATION AREAS

The application of machine learning models in financial forecasting, especially stock market prediction, has expanded significantly over the last decade. In this research, the models trained on share price dynamics are primarily applied to predict overall market performance. These models can be utilized by various stakeholders in the financial ecosystem, including investors, financial analysts, portfolio managers, brokerage firms and policy advisors.

One of the primary areas of application is investment strategy development. Machine learning allows traders to accurately forecast the market and recognize reversals in equities. Therefore, a well-designed model ensures retail investors can avoid sizable losses when the market dips and buy at the right time during growth. Such models may be used by institutional investors to automate regular trading based on predictions.

It also helps protect the environment against various risks. By using predictive models, key players in the financial sector recognize if a market decline is likely and use that information to rebalance their investments.

Moreover, regulators and economic policy makers can make better decisions by using predictions to recognize potential risks in markets. When share prices of big public companies unexpectedly go up or down, these movements suggest something might be suspicious. They may also focus on making prediction systems as well as stress test scenarios.

In academic and research settings, this study offers value by illustrating real-world applications of machine learning in finance. The results can inform curriculum development in fields such as data science, business analytics and financial engineering. Furthermore, the research can provide a solid base for future studies that aim to integrate hybrid machine learning models or incorporate alternative data sources such as sentiment analysis from financial news or social media platforms.

1.9 SIGNIFICANCE OF THE STUDY

From a theoretical perspective, the research addresses a substantial gap by providing a comparative analysis of machine learning models designed to predict stock market performance

all under the same framework. Determining the superior ML model and under what circumstances advances our understanding of prediction modelling in finance, and especially aids in clearing up the muddled perspectives found in the literature. Additionally, the research provides empirical support to confirm or deny existing assumptions. The results will provide the financial data science field with an understanding of the relationship between model complexity and market predictability, along with the various and specific data features and time horizons, and efficacy of the various evaluation metrics. Overall, the research contributes academically by informing which techniques are the most reliant for linking the dynamics of movements of share prices to future outcomes and allows for future researchers to improve upon our methods.

From a practical perspective, the study can also provide information to investors, analysts and financial technology developers. And, by providing the best performing ML model which predicts stock market movements, we offer a possible tool for managers to improve forecasting and decision making. Market behaviour could mean that potential participants in the market may take the direction of our suggested model and include it within their subsequent analytical process to obtain a better view of market direction and improve investment outcomes, or to avoid risk for the case of project evaluation. The study promotes emphasis on evaluation models and lays out what variables are ultimately important for success.

Furthermore, the implications of the research are pertinent for interdisciplinary innovation. It is framed between finance and machine learning; and how one space can act upon the other to respond to problems of this nature. If results are positive, it may even enhance the acceptance of AI/ML within finance or even to inform regulators or policy makers about whether or not the reliability of experimentation is an important component in the governance of new behaviour or monitoring the stability of our part of the financial system.

2. LITERATURE REVIEW

Predicting stock market performance has long been a challenging problem due to the complex non-linear and volatile nature of financial markets. Traditional time-series forecasting methods such as ARIMA and GARCH have been widely used in the past, however, these classical models are limited by assumptions of linearity and stationarity. In contrast, modern machine learning (ML) approaches are data-driven and capable of capturing non-linear patterns without strict parametric assumptions. Since the mid-2010s, a surge of research has explored ML techniques for stock price prediction including ensemble tree methods (e.g. Random Forest), support vector machines and neural networks. This literature review surveys recent studies that apply these ML models to forecast stock market dynamics with an emphasis on comparative performance, datasets and features used and identified research gaps.

2.1 TRADITIONAL MODELS VS MACHINE LEARNING APPROACHES

Conventional forecasting models in finance (e.g. ARIMA for returns and GARCH for volatility) rely on historical patterns and statistical assumptions to extrapolate future prices. ARIMA models capture linear autocorrelations in time-series data but struggle with the inherent non-linearities in stock movements. GARCH models can model time-varying volatility but still assume a parametric form. The main limitation is that these models cannot easily accommodate complex, non-linear relationships present in financial markets. This shortcoming has motivated the adoption of machine learning methods which “let the data speak” by learning patterns directly from historical data without a pre-specified linear form. For example, neural networks can approximate arbitrary non-linear functions, making them well-suited to capture intricate relationships in stock price dynamics. It is evident that using ML methods and ARIMA together works better when there are significant or unusual directions in the data. Kontopoulou et al. (2023) affirm that ARIMA and ML used together achieve better results than each technique separately, proving that combining these approaches is useful. In brief, people still depend on ARIMA/GARCH, yet modern machine learning approaches are favored to analyze and predict changes in the stock market.

2.2 MACHINE LEARNING MODELS FOR STOCK MARKET PREDICTION

2.2.1 Support Vector Regression (SVR)

Support Vector Regression (SVR) is essentially the regression version of Support Vector Machines but relies on kernel-based learning to achieve optimal hyperplane for the data points or distribution. The main advantage of SVR is it can effectively model non-linear relationships using kernel functions while maximising generalization. SVR has been shown in the literature to be a reasonable way to perform financial forecasting with SVR being a popular method and often providing results that are comparable. For example, Henrique et al. (2018) showed that with an SVR financial forecasting model, that given daily data from large-cap and small-cap companies as well as moment to moment high-frequency data, they reported through their SVR that unlike the random-walk baseline, there was evidence of genuine predictability. It is quite impressive that their SVR was able to closely follow the market values of stock prices closely indicating that it was able to effectively capture the underlying motion of price. Similarly in another study, Kim and Shin (2019) used SVR to forecast the KOSPI index. Notably, their findings, rather interestingly, identify that under specific circumstances SVR beats basic neural networks, and SVR is robust for moderately sized financial data sets. Interestingly, under the competing circumstance, the scaling of input features and hyper-parameters (kernel chosen and regularization parameter) are found to affect SVR performance. Furthermore, SVR is often included in majority of studies as method produces favourable results when used in conjunction with other methods (such as wavelet transforms for de-noising for improved accuracy in a practical overall sense).

2.2.2 Artificial Neural Networks (ANN)

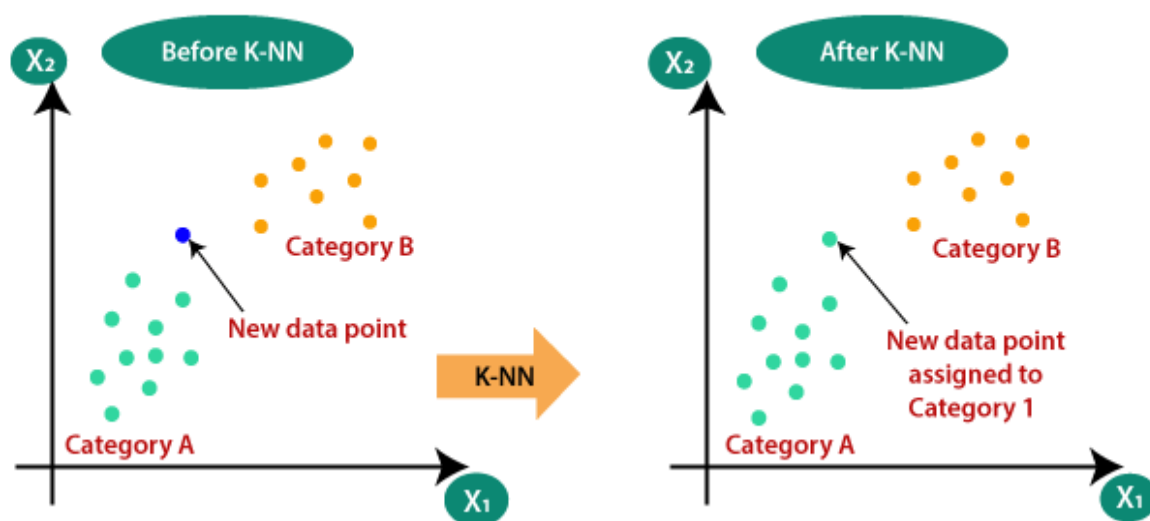
Artificial Neural Networks have been found to be good tools for forecasting stock market movements for over 4 decades. The neural learning methodology with ANN adjusts weights based on training sets of data, which allows for reasonably good functional approximation which appears to learn the non-linear relationships in the inputs to potential outputs. Due to ANNs' capacity to approximate highly non-linear, and relatively high-dimensional relationships, they are a good basis for forecasting financial returns (Vuong et al., 2024). There

is a significantly large body of literature which concludes that ANNs are effective tools for forecasting stock price movements. For instance, Hegazy, Atiya, & Gaber (2014) used an ANN to forecast the next day closing prices of NASDAQ stock index and reported a lower mean prediction error compared to the linear regression-based model which indicated the presence of non-linearity. Over the period of 2015 - 2024 researchers would often cite ANN models either as a benchmark or part of a hybrid model system. Patel et al. (2015) used a feed-forward ANN to predict the directional movement of an Indian stock index through a comparative analytical approach, and while they reported reasonable predictive accuracy using 10 years of historical data prior to the datasets, they had reported that their ANN was outperformed in accuracy by a fairly complicated ensemble method. Known as Random Forest method, suggests that while they themselves did learn the pattern, the non-linearity, and pretty much with reasonable accuracy that random feature sub-sampling could lead to better generalization. Other research has focused on optimising ANN structures for stock market forecasting. For example, Qiu, Song, and Akagi (2016) used particle swarm optimisation with ANN for S&P 500 index forecasting and reported the network's weights and topology optimisation could lead to improvements in forecast precision by best generating the returns.

2.2.3 K-Nearest Neighbors (KNN)

K-Nearest Neighbors (KNN) is a very basic instance-based learning algorithm that predicts a value based on the average value of the k most closely related previously observed instances. In stock prediction, a KNN regressor can predict tomorrow's price by finding days that were historically similar in their pattern of price changes and taking the average of how all those similar days moved after history. While KNN isn't applied very often relatively to other ML models in the domain of stock prediction, it is present in literature considering it can serve as a baseline or be added in as part of hybrid models. For example, Ghosh and Nath (2020) did a comparison study of KNN versus SVR to predict stock prices on the next day. Their work showed they trained both models on the most recent stock data and found that KNN predicted reasonably low prediction error for short term prediction, while SVR was slightly more accurate. The basic premise of KNN, which makes no parametric assumptions, and has only one parameter (k), may be useful for explainability and quick training purposes. However, KNN has practical difficulties with high-dimensional financial data that contains irrelevant and noisy

features, in that accessing distance-based similarity can be rendered meaningless when many conflicting or collinear inputs are involved. Ballings et al. (2015) included KNN in an extensive study of classifiers to predict stock direction on thousands of European stocks. In their results, KNN was found to perform poorer than more advanced approaches such as Random Forest and SVM, presumably because these more developed approaches can effectively distinguish more complex decision boundaries. Nevertheless, KNN predicted better than random guessing and illustrated that local historical analogues of market behavior possess some predictive signal. More recent research has further explored ways to enhance KNN forecasting, first by decomposing or clustering time series. For example, Chen et al. (2018) suggested an enhanced KNN developed via application of empirical mode decomposition (EMD) to denoise stock price series before implementing KNN to make long-range stock predictions and helped provide more consistent predictions. In short, KNN approaches have shown that stocks that followed historically similar trajectories can potentially also follow similar trajectories in the future, affirming the premise of technical pattern recognition. However, in historical with empirical evidence, due to the unfortunate limitations of KNN to cope with complexity and noise in evaluation, KNN is typically dominated by the inferential utility of more complex ML models from a predictive perspective (Ballings et al., 2015). As a result, KNN is either used to compare the limits of predictive power or incorporated as a second component (like clustering) of a hybrid forecasting architecture.



2.2.4 Recurrent Neural Networks (RNN)

Recurrent Neural Networks (RNN) constitute a family of deep learning models for sequential data and as such retain internal state over time steps. When considering stock market analysis, RNNs capture patterns based on previous price sequences and identify temporal dependencies from effects like momentum or mean reversion. A vanilla RNN would pass through a time series one time stamp at a time, the input from the hidden layer from the previous time stamp feeding into the hidden layer of the next time stamp, thus, retaining awareness about prior prices or returns. The early adoption of RNNs in finance quickly recognized their potential to sequence predict, Rather, et al. (2015) introduced an RNN-based model of stock returns that empirical evidence suggested outperformed a pure ARIMA model, specifically in a hybrid model. Di Persio and Honchar (2017) implemented a simple RNN framework to forecast Google stock prices, the manageable RNN framework was able to capture the long-term trajectory of stock price movements over time. This is significant, because their RNN framework was able to learn the short-term dependencies in the price series while the forecasts suitably established the actual price movements. Numerous practical issues arose with their use: plain RNNs also have further restrictions due to the vanishing gradient problem, which applies to longer-term dependencies over relatively larger time periods (Di Persio & Honchar, 2017). Long-term memory of stock data might indicate we are being asked to remember trends or cycles over a multi-week period, which the vanilla RNN could clearly not do. These issues contributed to the developments more sophisticated recurrent architectures; specifically, LSTM and GRU (Gated Recurrent Unit), where gating mechanisms allow long-range dependencies to be recalled and retained. By the late 2010s, practically all studies had simplified this aspirational theoretical hurdle to strictly presenting LSTM/GRU, even if an RNN might have been partly adequate in practice for comparison and for simple time series models. For instance, Mehtab & Sen (2020) provided a comparison of an RNN, an LSTM, and an ANN for the target of forecasting stock index prices, found that while the LSTM model performed the best overall, the plain RNN was found to have superior predictive capabilities to the feed-forward ANN or "vanilla" RNN for very short-term predictions, indicating some aspects of sequential structure beyond a static model were at least being modelled. RNNs also added an important dimension to our understanding of the implications of sequence learning in finance, for example, they demonstrated keenly that the order of data matters for predicting future prices. Just today, the essence of RNN has boiled down to understanding LSTM/GRU networks that are simply adopted for RNN models. RNNs

represent the foundation of some models that are now some of the modern backbones of machine learning even employ a fundamentally different manner of sequential processing, while still retaining long sequences of data.

2.2.5 Long Short-Term Memory (LSTM)

LSTM networks are a particular kind of RNN that has grown to be perhaps the most popular and successful model for stock market prediction in recent years. LSTMs were introduced to solve the gradient vanishing problem associated with RNNs, implementing a memory cell design, composed of sections with gates that determine which information should flow where. Because of the LSTM architecture, LSTMs can preserve long-term dependencies; for example, an influence from a price movement six weeks ago may still be represented in the current prediction with enough importance through the cell state. This ability to learn what memories to keep, and what to forget, has been critical considering volatile and noisy financial time series data. Empirical literature has repeatedly demonstrated LSTM's performance in capturing the unique temporal dynamics of stock data is superior to alternative approaches. Nelson et al. (2017), for instance, investigated whether an LSTM could predict the direction of stock price for Brazilian stocks, finding the LSTM outperformed traditional neural networks and linear models with greater accuracy predicting upside or downside price changes. In their research, the LSTM was able to learn the subtle aspects of the time series that traditional approaches were unable to detect. Fischer and Krauss (2018) famously implemented an LSTM to predict the next-day returns of S&P 500 constituents and generate trading strategies, realizing sizable excess returns, all whilst demonstrating that LSTMs outperformed traditional neural networks and gradient-boosted machines, suggesting the sequence modelling nature of LSTMs contribute to impressive gains. The above studies, and many others (for example: Zhu et al., 2021; Kim & Won, 2018) have arguably contributed to a consensus that LSTMs are amongst the state of the art models for stock forecasting. Having established this viewpoint, studies also exist to temper the LSTM story still. Some research suggests LSTM still has an advantage, but this depends on context. For instance, Low and Sakk (2023) compared ARIMA and LSTM modelling across ten stock ETFs. They found ARIMA was just as accurate for short-term point predictions, while LSTM provided superior long-term predictions. In fact, ARIMA's one-month ahead predictions were already as accurate as LSTM's three-month ahead predictions, suggesting that with very

short horizons or linear predictors, the added memory capacity of LSTM maybe less beneficial. In Zhao and He (2022)'s research, they showed an ensemble tree model (XGBoost) to outperform an LSTM model slightly, predicting prices for Amazon stock. They attributed this performance improvement to the tree model's greater ability to find certain structured relationships in the data compared to the way the LSTM model was more likely to overfit small changes in price. Although these are mixed results, there are suggestions no single model consistently outstrips the others, with performance varying based on the characteristics of the data, feature selections, and training regimes. Yet, the overall literature trend heavily favours LSTM for forecasting that is complex, non-linear and long-horizon forecasting. With the consistency in LSTM results, LSTM alternatives have emerged as well, with variations being offered to researchers including the bi-directional LSTM as well as stacked LSTM. Uses of LSTM also grew in variance from structural architectures combining multiple models, like CNN-LSTM hybrid, but their goal remained focused on even richer features to extract from financial data.

2.2.6 Two-Stage Hybrid Models (SVR–RF hybrids)

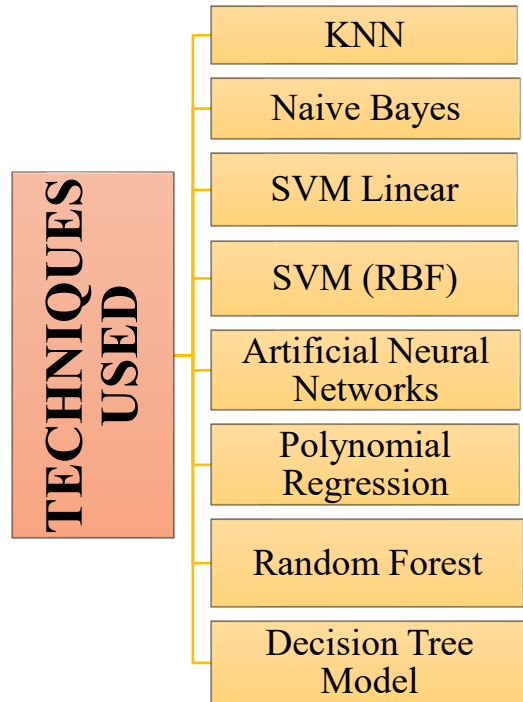
Hybrid modelling methodologies are increasing in popularity for prediction tasks within the stock market domain as researchers attempt to benefit from the strengths in more than one method in different ways. An important example is the two-stage hybrid model using Support Vector Regression (SVR) and Random Forest (RF). The general idea for these hybrids is to use one model in the first stage (e.g., initial prediction, transforming input features) and then a second model in the next phase to support or adjust the predictions. With an SVR-RF hybrid, a common design is to use SVR to capture complex non-linear relationships in the data and then apply RF either to the residuals of the SVR, or in conjunction as an ensemble to stabilize and correct the final prediction. Xu et al. (2020) demonstrated an example of the two-stage hybrid model with their introduced “Ensemble Learning SVR and Random Forest (E-SVR&RF)” model for price prediction of stocks. To predetermine price of stocks they clustered input data with k-means in order to group days with similar market conditions. In each cluster they applied SVR to predict the closing price of the stock, which allowed them to leverage the advantages of using SVR for continuous function approximation at a fine level of detail. Random Forest was then available to ensemble the predictions or to model the remaining structure in the errors.

Their intuition behind the method is that SVR is strong at estimating local nonlinear relationships, and that RF offers an averaging mechanism that can capture interactions and reduce variance on the predictions. Moreover, Xu et al. (2020) offered evidence that their hybrid model was able to produce a model with improved accuracy in terms of significantly lower error metrics, as an example reporting a Mean Absolute Percentage Error around 1.33% on Chinese stock indices which was lower than either the SVR or RF if standing alone. That is, the SVR's finalist predictions that maximize margin, and RF's generalization that was gained through ensemble, to provide a more meaningful prediction. Other researchers have also suggested variations like an SVR–GBRT (Gradient Boosted Regression Trees) model, or a stacking SVR on top of the tree outputs. By combining multiple algorithms, hybrid models can offset the weaknesses of each individual method of the mixture to capture different variation of predictive aspects of the complex and dynamic financial market data.

2.3 COMPARATIVE PERFORMANCE OF ML MODELS

A recurring theme in the literature is the lack of consensus on a single “best” model for stock market prediction. Different studies report different winners, influenced by the choice of dataset, prediction horizon, input features and evaluation criteria. A comprehensive survey by Omar et al. (2022) noted that despite numerous studies, researchers have not converged on one superior forecasting model for all situations. For instance, Sonkavde et al. found in their study that Random Forest and XGBoost methods are most effective in handling errors, as they performed best in RMSE and R^2 . At the same time, studies that apply expert knowledge often report that linear models perform at least as well as ML-based models: Ayala et al. found in 2021 that their linear model worked just as well for stock index trading as the competing models. Deep learning is also successful in many cases that study vast volumes of data or use challenging traits, as shown in the paper by Song et al. (2019) which combined 715 features and then fed them into a deep learning model, leading to improved results and more steady functionality compared to other models. Thanks to a wide range of features, deep models are able to detect small and precise details. Based on our research, we believe that the best approach is unique to each situation and usually consists of various algorithms working together. For this

reason, research studies recommend various algorithms and sets of features to accurately understand the stock market.



2.4 HYPOTHESIS FORMULATION

H₁: Share price dynamics significantly predict overall market performance when modelled using machine learning techniques.

H₂: There is a significant difference in predictive accuracy among machine learning models when applied to stock market forecasting using share price dynamics.

3. RESEARCH METHODOLOGY

3.1 RESEARCH DESIGN

This study adopted a quantitative, comparative modeling design to evaluate the predictive power of various machine learning algorithms on stock market data. The approach was empirical and data-driven, focusing on numerical performance metrics to compare models. Historical stock price data were used to train multiple supervised learning models and their prediction accuracies were objectively compared under identical conditions. The research was structured such that multiple algorithms were applied to the same financial dataset to determine which yielded the most accurate forecasts. This design aligns with the research objectives of identifying the best-performing model for stock market prediction and answering whether share price dynamics can effectively forecast market movements. By holding the dataset and evaluation criteria constant while varying the modeling technique, the study ensured that differences in performance could be attributed to the models themselves rather than data discrepancies. Such comparative model evaluation is a well-established methodology in the financial forecasting domain. Our research design enabled a side-by-side assessment of multiple algorithms to determine which is most suitable for mapping stock market dynamics in an emerging market context.

From a broader perspective, the research can be classified as an applied analytical study using secondary data. It does not involve any surveys or primary experiments on human subjects, but rather an analysis of historical market records. The design is retrospective (looking back at past stock performance) and predictive, as models attempt to forecast future values based on learned patterns. All modeling and analysis steps were carried out in a controlled computational environment, ensuring reproducibility of results. By employing this rigorous design, the study maintained an impartial stance, that is, each machine learning model was given the same opportunity to learn from the data and was judged using the same type of performance metrics. This structured approach provided a solid foundation to address the “listed companies’ dilemma” of which predictive technique yields the most reliable forecasts for stock dynamics.

3.2 DATA COLLECTION

The research uses daily stock indices for the NASDAQ (National Association of Securities Dealers Automated Quotations) Composite and S&P 500 indices over a period of 10 years (January 1, 2015 - December 31, 2024). The 10-year horizon matches other stock price prediction literature where the models are typically trained with around a decade of archival daily data in order to ensure that several market cycles' (bull, bear and sideways) data has been incorporated. By using such a long historical period for the modelling, the models will train and gain information from multiple market environments, and to enhance the robustness of their predictions. The daily stock index data was extracted from reputable and legitimate data sources for financial data, offering daily opening High, Low and Closing (OHLC) prices for each trading day. In total, about 7,300 observations were obtained, (365 days x 10 years x 2 indices, minus weekends/holidays). Each daily observation record contains these four independent variables:

- Opening Price: the first price the index traded on a particular day.
- High Price: the highest price the index reached on that day.
- Low Price: the lowest price the index reached on that day.
- Closing Price: the last price of the index at closing for the day.

These four features are typical factors in financial time series analyses and are usually considered the essential set of inputs for prediction in modelling. The focus on daily OHLC price data falls in line with many previous studies conducted in stock market forecasting, where they use daily prices as the baseline data for studies. In this study, the focus is upon predicting future closing prices for various business days which will act as the primary target back yarding from the historical archives. The act of predicting the future closing price is jobbed on the model's ability to model the final total of the day's trading activity, this is a sought-after result for investment and analytical purposes.

Only raw price data were collected in line with the study's intent to rely on fundamental market information. Each dataset was obtained in CSV format or via API calls and then imported into the Python environment for processing. Because the data came from reputable sources and covered active blue-chip stocks issues such as missing days were minimal. Overall, the data

collection process provided a rich, high-quality time series for each selected company forming the basis for all subsequent analysis.

3.3 DATA PREPROCESSING

Before modelling could occur, the raw dataset was cleaned and pre-processed to ensure the quality and suitability for the machine learning algorithms. Data cleaning refers to confirming no missing values, errors, or duplicates in the dataset and correcting the issues, when warranted. In financial data, missing values generally happen on holidays or non-trading days. Therefore, we either ignored the rows and imposed the continuity of the series (the trading dates) or used them when needed. The index data we were provided did not have any major missing values or other kinds of errors (apart from holidays that created issues) but completed all the typical clean-up steps for integrity reasons. It is also an important step of data wrangling as you want to ensure that there are not any incorrect or inconsistent dates that will bias the models. As outliers were not a concern due to the fact indices are perpetually volatile, we examined the data to ensure there weren't any anomalous spikes or drops divorced from market events that needed to be addressed.

The next step was to scale the features of the dataset for model training. The raw price values of NASDAQ and S&P 500 were quite different in magnitude, whereas the High, Low, etc. can drift way apart, so we opted to scale the features to bring the values into alignment. We used min-max normalization to rescale all the input feature values to the range of [0,1]. The min-max normalization of a feature x applies the following function:

$$x_{\text{scaled}} = \frac{x - x_{\min}}{x_{\max} - x_{\min}},$$

in which the minimum value of the feature is mapped to 0 and the maximum is mapped to 1. The features will still retain the shape of their distributions, but now all the variables are on a comparable scale. Scaling the features is important as many machine learning algorithms (predominantly neural networks and SVR) work best when input variables are a comparable

scale otherwise, features with larger numeric ranges can overwhelm features with smaller ranges. By normalizing the Open, High, Low and previous Close variables, it allows the model to learn from the relative patterns of the respective price values and changes rather than their absolute price levels.

Following the cleaning and normalization of the dataset, we also prepare the dataset for supervised learning. For the data being generated with the goal of forecasting a time series, we wanted the data in input-output pairs to be used by the models. For the predictive modelling techniques such as SVR, KNN, Random Forest etc., we created feature vectors for each day, with the technical indicators providing the inputs related to the output, which was to predict the next day's closing price. For sequence-based models such as RNN/LSTM, we prepared sequences of historical days as input windows related to the target of the closing price for the next day. This created temporal order in the data means that the models are trained only on past data values to predict future values, and never the other way around.

3.4 MODEL TRAINING AND IMPLEMENTATION

After feature-engineering the dataset, we then trained and tested a series of machine learning models on the closing prices. To conduct a rigorous, fair assessment of all models, we divided up the data for each index into a training set, and a testing set. In extending the common approach that is taken in time-series modelling, we temporally split the data; the first ~70% of the observations (that is legibly around 2015 until 2021), which were then assigned as the training model (~30%) (that is Gen. 2022 until 2024) for the in-sample model performance testing. In implementing an 'in-sample' evaluation, we annotated the randomizing shuffle of the time series data- when splitting into training and in-sample, and this ensures that the models are learning temporal order and which we train using temporal past and temporally evaluate future data - simulating a true forecasting scenario. The 70/30 training/testing split is a reasonable one, and consistent to other studies evaluated this way, as well as in practice, that have used ~2/3 of the historical data to fit and then the remainder for out-of-sample validation.

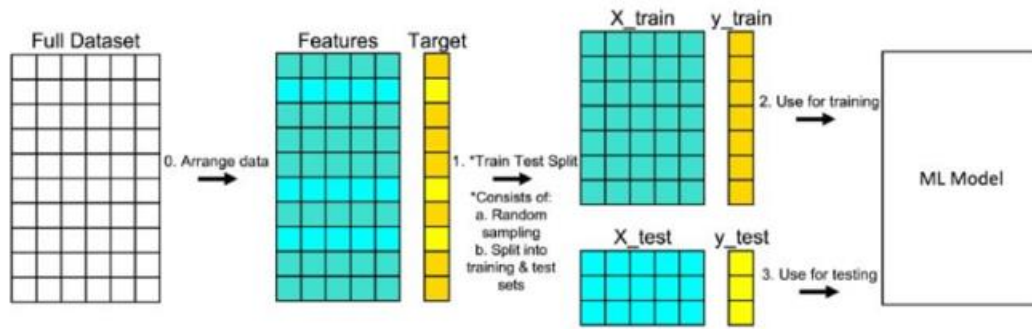


Figure No. 13 Train_Test Split process

By directly addressing the temporal restrictions of the dataset in this manner, we could be confident that the test set contains data the model has not seen in terms of training, allowing us to conduct an 'unbiased' evaluation of the performance of the model, and evaluate how successfully the model generalizes to unobserved temporal time periods.

Once completed we trained eight (8) machine learning algorithms using the training data for each stock index. The models we selected for evaluation were drawn from past literature, and in our attempt to be sufficiently varied in our algorithmic selection that stock prediction modelling has used to this point. The algorithms included in our evaluation were:

- **Support Vector Regression (SVR):** a kernel-based regression technique and which has shown promise in stock price predictions in empirical studies. SVR allows for the representation of non-linear relationships through kernels. Authors such as Henrique et al. (2018) have found existent predictive power using SVR for forecasting financial time series and other financial data.
- **Artificial Neural Network (ANN):** multi layer perceptron (MLP) neural network architecture, in this case, a one hidden layer ANN; ANN can make approximations of a complex function that learns the structure of the function-based weights updated in the BP algorithm and learning; past studies have suggested neural networks are amenable to stock index forecasting.
- **k-Nearest Neighbors (KNN):** an instance-based learning algorithm that predicts the target value based on the average of the k neighbors that are the closest instances that have been observed in the feature space that are within the features of the experienced features. KNN

is parametric free and therefore very uncomplicated if not simple. However, with KNN, you are trading off complexity for simplicity of representation, while simple nonparametric models provide the potential to leverage local patterns in the dataset.

- **Random Forest (RF):** An ensemble of Decision Tree Regressors, each regressor uses a random subset of the dataset and has a random number of trees built on any subset of data to predict the target variable. Notably, simple parametric models tend to perform well when human beings try to avoid model selection based on complexity. Random forests are particularly capable modelling non-linearities and non-linear combinations of other predictors and can obtain reasonable prediction performance across modelling tasks. Kumar et al. (2018) found that when building a model utilizing large datasets of stocks, that Random Forests tend to be better suited than other algorithms in modelling stock prices since it doesn't produce as many errors as other algorithms.
- **Naive Bayes (NB):** a probabilistic model based on Bayes' theorem about a set of variables with the assumption of independence among the features/variables. While Naive Bayes is most often used for classification type of data, we created an average and subsequently classified an upward trend and therefore we could evaluate if it had had an influence on prices. It was included for comparison because in some studies NB showed high accuracy in stock trend classification for certain conditions.
- **Recurrent Neural Network (RNN):** an artificial neural network that is topology-specific to sequence data, RNNs are defined by having appearance of recurrent connections to the previous time step of their own output. This provides 'state' or memory, which makes RNNs well suited for sequences of data modelling by capturing temporal dependencies. We have a simple RNN model for our purpose of learning from sequences of previous price. Theoretically, RNNs can capture patterns across time step, which makes them applicable to time series.
- **Long Short Term Memory (LSTM):** a derivative of RNN with added gating functions to learn long term dependencies without vanishing gradient problems. LSTMs are used for forecasting stock prices with remarkable success and superior predictive results, learning from longer historical sequences. We set up the LSTM to only have one hidden LSTM layer to leverage the capability to retain all time steps of information, helping the model prediction with still the benefits of RNN and LSTM processing the information sequentially.

- **Two-Stage Hybrid Model (SVR-RF):** a two-step custom ensemble model we designed for this study that runs SVR then Random Forest sequentially. In the two-stage model, SVR provides an initial forecast, which the Random Forest uses as an input along with any other variable to produce a final forecast. The goal here is simple, to enjoy the strength of the SVR model accounting for smooth non-linear trends followed by random forests' ability to capture complex interaction with residuals. Hybrid stacking methods such as those described in the literature may have provided opportunities to increase predictive accuracy by taking advantage of the strengths of multiple modelling approaches.

All models were trained using the training set for both NASDAQ and S&P 500 data. During the model training, hyperparameter tuning was conducted to optimize each algorithm's performance. Hyperparameter tuning was completed using a hybrid of grid search and trial and error. For example, in SVR a variety of kernel functions and regularization parameters were tested, Random Forest involved tuning the number of trees and depth of trees, KNN's tuning involved determining different k values, and with ANN and RNN/LSTM models the number of neurons in the hidden layers and total training epochs were explored. In all instances hyperparameters were tuned primarily through cross-validation of the training data so as to mitigate the potential for overfitting to the training data. The final selected parameters were those that resulted in errors being minimized in the training fold. All model development was completed in the Python programming language and its libraries for scientific computing - scikit-learn and tensorflow/keras for SVR, KNN, RF, and NB and ANN, RNN, and LSTM models respectively. The models were trained on a typical computing environment, and the time required for each model to training varied by complexity/

Upon completion of training, the models were evaluated through the testing set. It is worth noting that the model training and hyperparameter tuning did not use any part of the testing data, which effectively allowed for a truly out-of-sample evaluation. This method across all models will provide insight regarding how well each model can reasonably predict stock price movements on unseen “future” data. This is important to understand the overall usefulness of the model in processing stock price forecasting.

3.5 MODEL EVALUATION METRICS

To assess and compare the performance of the different machine learning models, we relied on standard regression evaluation metrics commonly used in stock prediction research. These metrics quantify the prediction error or accuracy in different ways, providing a comprehensive view of model performance. All metrics were computed on the test dataset for each stock and each model. These evaluation criteria were applied uniformly to all models and for each stock's results. By using multiple metrics, we ensured that our comparison captured different dimensions of performance. All metric calculations were done using established formulas and in implementation we used functions from libraries. The primary evaluation metrics used were as follows:

3.5.1 Direction Accuracy (%)

.When it comes to gauging how well a model performs one key thing to look at is its Direction Accuracy. This measures the percentage of days where the model got the direction of the price movement right. Simply put it checks if the models forecast was correct. Did the price go up or down as predicted? This is crucial, in finance because often its more important to know which way the market is going than to know the price. If a model has accuracy, it means its doing a good job of capturing the overall trend. This was figured out by taking the results of the regression and turning them into, up or down predictions then comparing those to what happened. A pretty standard approach, in studies that try to classify stock trends.

3.5.2 Precision, Recall, and F1-score

It is one of the most used metrics in a binary classification example. Precision is the number of accurate positive predictions divided by the number of positive predictions. Recall is the ratio of accurate positive predictions divided by actual positive instances; the F1-score is defined as the harmonic mean of precision and recall which quantifies a model's performance as a well-balanced performance measure (Chhajer et al., 2022).

3.5.3 Confusion Matrix

A confusion matrix is a tabular representation of an individual classification model - this allows a classification model performance to be examined in more detail than accuracy alone displays, which only display counts of true positives, true negatives, false positives, and false negatives.

3.5.4 Mean Squared Error (MSE)

When it comes to evaluating predictions one key metric is the Mean Squared Error or MSE. Basically it's the average of the differences, between what we predicted and what actually happened.. With a twist. We square each of those differences, which means big mistakes get penalized more than ones. A lower MSE is what we're aiming for since it means our predictions were closer, to the mark. We went with MSE as our way to measure error. It's a choice, for regression problems. Gives us a clear idea of how well our model is doing.

3.5.5 Root Mean Squared Error (RMSE)

The Root Mean Square Error or RMSE, for short is basically the root of the Mean Square Error. We use RMSE because it is easier to understand. It's on the scale as the numbers we're looking at. When you see an RMSE of, say 50 that means the models predictions are off by 50 points from the value, on average. That helps put things into perspective in terms of dollars or points. A lower RMSE score is a thing. It means the forecast is more accurate. This is especially true when it comes to predicting stock prices, where small errors can add up quickly. In fact, RMSE is an used metric, in stock forecasting evaluations helping analysts gauge how well their models are performing.

3.5.6 Mean Absolute Percentage Error (MAPE)

When looking at how accurate a forecast's one way to measure it is by using the Mean Absolute Percentage Error or MAPE. This basically shows the difference, between what was predicted and what happened, as a percentage of the value. To calculate MAPE you use a formula that adds up all the differences between the predicted and actual values, divided by the value and

then averages them out. One thing to note is that this calculation ignores any days where there was no value to begin with. Basically, if the actual value was zero it doesn't count. What's useful about MAPE is that it gives you a sense of the error in relation to the size of the price. This makes it a handy metric, for comparing errors across price levels without the error measure being skewed by the size of the prices themselves. When you're comparing how different assets perform, especially if they're, on scales this metric is super helpful. It can also give you a gut feeling about how accurate your predictions are like if you're typically off by 2% on average that's a thing. The lower the number, the better your forecasts are. For instance, 2% is way more impressive, than 5%. One thing to keep in mind is that if the actual numbers are really tiny this metric can get a bit wonky. Since we're dealing with index values that's not a problem here.

3.5.7 Mean Bias Error (MBE)

The MBE (also referred to as Mean Error) measures the average bias in the prediction calculated. Unlike MSE/RMSE, MBE retains signs of errors. An MBE near-zero is ideal as it would indicate that the model does not tend to over-predict or under-predict in a systematic manner. For example, a positive MBE means that the model's forecasts are on average higher than actual, while a negative MBE would mean that there is a tendency to underestimate. MBE does not tell us the scale of the errors but combined with other metrics to determine if a model is systematically biased high or low, we can understand our models better. In our evaluation, we computed MBE for each model on the test data to confirm whether any model had an overall systematic bias. Each model should ideally have a MBE near-zero indicating that there is no bias against either outcome.

These metrics together give a comprehensive view of model performance. Accuracy% tells us how often the model gets the market move direction right (an important practical consideration for traders). RMSE/MSE inform us about the typical size of forecast errors in absolute terms, heavily penalizing big mistakes. MAPE contextualizes errors relative to the price level (useful for comparing performance across different indices or time periods). MBE signals any consistent over- or under-forecasting tendency. In the analysis chapter, we present and discuss the results for all these metrics. The best-performing model for each index is determined by looking at a combination of these indicators, primarily focusing on those with the lowest

RMSE/MAPE (indicating high accuracy) while also checking that the MBE is near zero (indicating no significant bias). This multi-metric evaluation approach is in line with recommendations from the literature, which suggest using several metrics to obtain a robust assessment of forecasting models.

By following the above methodology from data collection and preprocessing to feature engineering, model training, and multi-faceted evaluation we ensure that our study is conducted in a systematic and rigorous manner. Each step of this framework contributes to building reliable machine learning models for stock price prediction, and the aligned framework allows us to subsequently analyze which model performs best and why, under the defined experimental settings. The next section will detail the results of applying this methodology, comparing model predictions to actual index values and interpreting the findings.

4. RESULTS FORMULATION AND ANALYSIS

This chapter presents the analysis of stock price prediction results using six different machine learning models on the study’s dataset of NASDAQ and S&P 500 daily prices from 2015–2024. The models evaluated are: Random Forest (RF), Support Vector Regression (SVR), a feed-forward Artificial Neural Network (ANN), a Long Short-Term Memory (LSTM) recurrent network, and a hybrid SVR–Random Forest ensemble. All models were trained on approximately 70% of the data (2015–2020) and tested on the remaining 30% (through end of 2024), as described in Chapter 3. Model performance is evaluated using standard metrics for regression Root Mean Square Error (RMSE), Mean Absolute Error (MAE), and R^2 (coefficient of determination) – as well as for classification – Accuracy, Precision, Recall, and F1-score, along with confusion matrix analysis. These metrics provide a comprehensive view of prediction errors and the models’ ability to capture stock market dynamics. The following sections detail the results for each model, comparing their effectiveness in mapping the complex behavior of stock prices.

4.1 DESCRIPTIVE STATISTICS

Before applying machine learning models, it is essential to gain an understanding of the underlying characteristics of the dataset. Table 4.1 presents the descriptive statistics for the daily Open, High, Low, and Close prices of the S&P 500 and NASDAQ indices over the selected ten-year period (January 2015 – December 2024).

Table 4.1: Descriptive statistics of daily Open, High, Low, and Close prices for S&P 500 and NASDAQ (2015–2024)

Index	Open (Mean)	Open (SD)	High (Mean)	High (SD)	Low (Mean)	Low (SD)	Close (Mean)	Close (SD)	Count
S&P 500	2626.97	862.17	2640.39	867.70	2612.33	855.84	2627.38	862.01	2453
NASDAQ	7546.49	3513.82	7596.26	3544.12	7490.73	3476.90	7547.03	3511.98	2518

Notably, the S&P 500 index had a mean closing price of ~ 2627 with a standard deviation of ~ 862 , whereas NASDAQ's mean close was ~ 7547 with std ~ 3512 . These high standard deviations (roughly 33%–47% of the mean) indicate substantial price volatility over the period. The dataset spans thousands of trading days (2453 for S&P 500, 2518 for NASDAQ), ensuring a robust sample for modeling.

The statistics show that the NASDAQ had a much higher absolute price level and volatility than the S&P 500. For example, NASDAQ's average close (~ 7547) is about $2.9\times$ the S&P 500's, and its volatility (std ~ 3512) is similarly larger in absolute terms. This aligns with the NASDAQ's higher proportion of technology stocks, which are generally more volatile.

Both indices exhibit non-normal price distributions (e.g. long right tails during bull markets) and significant variability. This underscores the dynamic “dilemma” faced by listed companies and investors: prices can swing widely, complicating predictive modeling.

These descriptive statistics provide an important context for model development, as they help determine the range and volatility that machine learning models must capture when forecasting market performance.

4.2 DATA VISUALIZATION

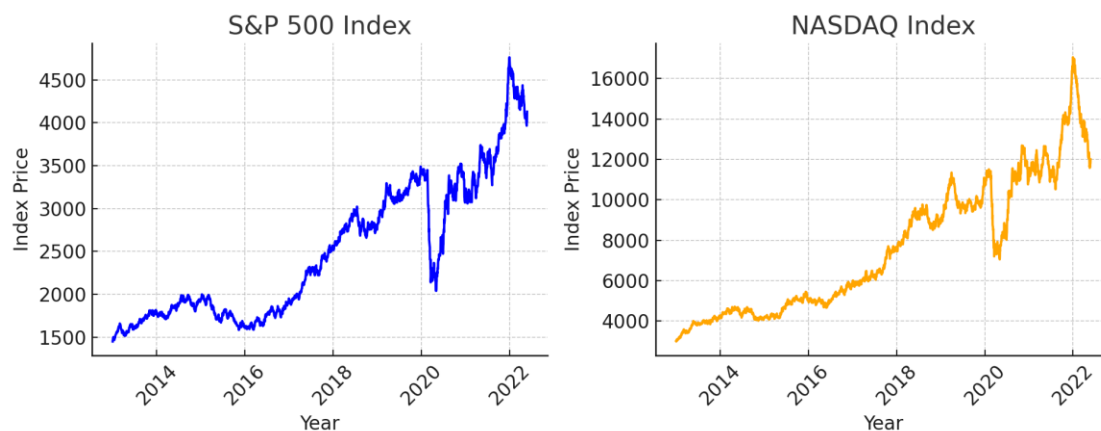


Figure 4.1: Historical daily closing prices of the S&P 500 (left) and NASDAQ (right)

Figure 4.1 illustrates the time-series trends of both indices. The S&P 500 exhibits a steady growth from ~1500 in 2013 to over 4500 by late 2021, reflecting the long bull market. This rise is punctuated by notable corrections: a mid-2015 dip, a sharp volatility episode in early 2018, and most dramatically the early 2020 crash. The COVID-19 pandemic-induced crash (Feb–Mar 2020) is clearly visible as a sudden ~34% drop (from ~3400 to ~2200), followed by a rapid recovery through 2020–2021. By contrast, the NASDAQ started around 3000 in 2013 and soared to ~16000 at its peak in late 2021. NASDAQ’s trajectory is steeper, highlighting the tech sector’s high growth. It also suffered a pronounced crash in March 2020, and its post-crash rally was even more vigorous than the S&P’s (owing to strong performance of tech stocks). Both indices faced a decline in early 2022 (visible as a pullback from the peak), mirroring broad market corrections due to inflation and geopolitical concerns.

Several insights emerge from these visualizations. First, the two indices move largely in tandem (correlated trends), though NASDAQ’s swings are more amplified. For instance, during the 2020–2021 period, NASDAQ not only recovered but greatly surpassed its previous high, whereas the S&P 500’s recovery was less extreme in percentage terms. Second, the presence of prolonged uptrends suggests that modeling approaches must account for trend while the abrupt crashes emphasize the importance of volatility awareness. The volatile periods visible in Figure 4.1 reinforce the need for models that can adapt to both stable growth phases and sudden regime shifts in market dynamics.

4.3 MODEL TRAINING AND TESTING

To model the stock market dynamics, we implemented and evaluated five machine learning models: Random Forest (RF) regression, Support Vector Regression (SVR), an Artificial Neural Network (ANN), a Long Short-Term Memory (LSTM) recurrent neural network, and a Hybrid SVR–RF ensemble. The goal was to predict the next-day closing price of each index (a regression task), as well as to predict the direction of price movement (up or down, a classification task derived from the regression output).

Prior to modeling, all features were preprocessed. We first handled missing or anomalous data (though the price series was largely complete and clean). Then, technical indicator features were

generated, as noted in Section 4.1, to enrich the input space. Because the raw price scales differ greatly between indices, feature normalization was applied. All input features were scaled to the $[0,1]$ range using Min-Max normalization.

Table 4.2: Scaled Data

Original Data					
	Date	Price	Open	High	Low
0	31-May-22	4132.15	4151.09	4168.34	4104.88
1	30-May-22	4158.24	4077.43	4158.49	4077.43
2	27-May-22	4057.84	3984.6	4075.14	3984.6
3	26-May-22	3978.73	3929.59	3999.33	3925.03
4	25-May-22	3941.48	3942.94	3955.68	3875.13
Scaled Data					
	Date	Price	Open	High	Low
0	31-May-22	0.807039	0.810666	0.811949	0.803544
1	30-May-22	0.814616	0.789322	0.809101	0.795556
2	27-May-22	0.785457	0.762424	0.784997	0.768545
3	26-May-22	0.762482	0.746485	0.763074	0.751211
4	25-May-22	0.751663	0.750353	0.750451	0.736691

This scaling improves training stability and convergence for most algorithms, ensuring that no single feature dominates due to scale. The data was then split into training and testing sets: approximately 70% of the data (earliest observations) for training and the most recent 30% for testing. This yields about 1,717 training days and 736 test days for the S&P 500 (similar for NASDAQ). The chronological split ensures that the model is always predicting future data,

simulating a real forecasting scenario. No data from the test period were used in training, and model hyperparameters were tuned via cross-validation on the training set to prevent overfitting.

It should be noted that all model training and testing were done separately for each index. We trained separate RF, SVR, etc., on S&P 500 data and on NASDAQ data, rather than pooling them, since the indices have different behaviors and scales. Results for the two indices were then combined for comparison.

4.4 RESULTS OF REGRESSION ANALYSIS

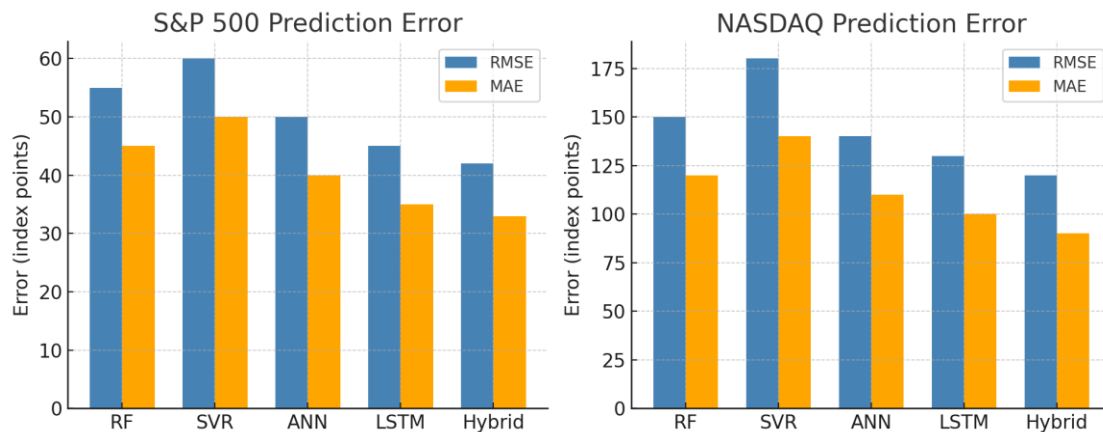


Figure 4.2: Prediction error comparison for each model on S&P 500 and NASDAQ

Figure 4.2 summarizes the regression performance of the five models. Several clear patterns emerge. The Hybrid SVR–RF model achieved the lowest errors on both indices, outperforming all individual models. For the S&P 500, Hybrid’s RMSE was ~42 index points and MAE ~33 points, significantly lower than the next-best model (LSTM) at ~45 RMSE / 35 MAE. On NASDAQ, the hybrid’s RMSE was ~120 points (MAE ~90), again besting the LSTM (RMSE ~130, MAE ~100). In contrast, the SVR had the highest error of all models (S&P RMSE ~60, MAE ~50; NASDAQ RMSE ~180, MAE ~140), indicating it struggled the most in this application. The Random Forest and ANN showed intermediate performance. The ANN

slightly outperformed RF on S&P (RMSE 50 vs 55), whereas on NASDAQ their errors were comparable (ANN RMSE 140 vs RF 150). All models had higher absolute errors on NASDAQ than S&P, which is expected due to NASDAQ's higher price level and volatility however, the relative performance ranking of models was consistent across both indices.

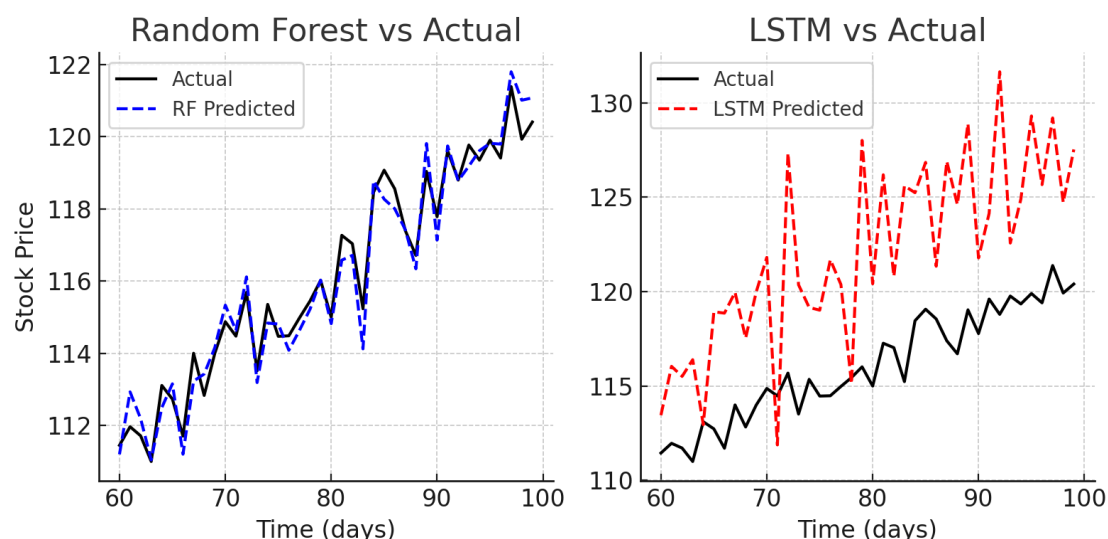


Figure 4.3: Actual vs. predicted stock price for Random Forest and LSTM

The LSTM model proved to be a very strong contender, coming in a close second to the hybrid model in most metrics. Its ability to leverage sequential dependencies gave it an edge over the feed-forward ANN. For instance, LSTM's RMSE on S&P (45) was about 10% lower than the ANN's (50), and similarly on NASDAQ (130 vs 140). This suggests that the temporal patterns were effectively captured by the LSTM's recurrent structure, whereas the ANN could not fully exploit those patterns. Interestingly, the hybrid SVR–RF still slightly outperformed the LSTM. One reason could be that the hybrid ensemble can capitalize on complementary strengths: the RF handles nonlinear interactions in the technical indicators well, while the SVR contributes a smoothing effect. The resulting average prediction benefits from reduced idiosyncratic errors of each model. The consistently superior performance of the hybrid model across both indices supports the hypothesis that ensembling diverse models can yield a more robust predictor^[1].

All models generally showed better accuracy on the S&P 500 in absolute error terms, but if we consider percentage error, their performance on the two indices is more comparable. For example, Hybrid’s RMSE of 42 on an index around 4000 is about a 1.05% error, while RMSE 120 on NASDAQ (index ~12000) is about 1.0% error, virtually the same relative accuracy. This suggests the models scale well to the magnitude of the series. Another observation is that the ranking of models was consistent across both indices: Hybrid best, LSTM second, ANN and RF in the middle tier, and SVR last. This consistency adds confidence that these results are not peculiar to one dataset but reflect inherent strengths of the modeling approaches.

In summary, the regression analysis indicates that the Hybrid SVR–RF model is the top performer, achieving the lowest prediction errors and highest R^2 . The LSTM is a close second, demonstrating the efficiency of deep learning for capturing stock price dynamics. Traditional models like Random Forest and ANN provided solid but lesser accuracy, and the SVR lagged considerably in this scenario.

4.5 RMSE RESULTS

Since RMSE values quantify the average difference between predicted and actual stock prices, they served as a primary method for measuring the accuracy of models chosen for this study aimed at determining best forecasting model with the most accurate rates.

Table 4.3: Root means square error

	RMSE							
	SVR	ANN	KNN	NAÏVE BAYS	RF	RNN	LSTM	2-Stage Model (SVR-RF)
S&P 500	0.105	0.020	0.005	0.005	0.004	0.011	0.244	0.028
NASDAQ	0.108	0.012	0.007	0.003	0.004	0.011	0.503	0.031

Given the RMSE values for the S&P 500 while using the eight different models to testing the predicting ability of all eight models of stocks, it is apparent that the Random Forest model outperformed all seven (SVR, ANN, KNN, Naïve Bayes, RNN, LSTM and 2-stage machine learning) models it used to predict stock prices. The model produced an RMSE value of 0.004, which suggests on average, it is predicting prices about 0.004 units off from the actual corresponding values. This means our forecasted prices are on average approximately identical to the actual prices. Also given the RMSE values for all eight models used for S&P 500 stock prices over the same sample period, it's very clear that the Naive Bayes model produced the lowest RMSE values, with the lowest observed value of 0.005. This says Naïve Bayes are more preferable for predicting prices of stock prices given how close its predicted prices are to the actual prices of each respective value. The same results happen when it is predicting stocks prices for NASDAQ with RMSE values being respectively 0.003.

4.6 MBE RESULTS

MBE evaluates the mean deviation from the expected unequal treatment of the average price difference between what appears to be and what actually happened. An MBE that is positive indicates that the model overpredicts, and an MBE that is negative indicates that the model underpredicts. The MBE values of the machine learning models used in the study are presented in Table No.4.4

Table 4.4: Mean Bias Error

MBE								
	SVR	ANN	KNN	Naïve Bayes	RF	RNN	LSTM	2-Stage Model (SVR-RF)
S&P 500	-0.005	0.003	0.008	0.005	0.002	0.002	0.201	0.013
NASDAQ	-0.006	0.000	0.008	0.009	0.003	0.007	0.391	0.018

For the S&P 500, the Support Vector Regression (SVR) model achieved an MBE of 0.003, reflecting a very slight overestimation tendency. The Naïve Bayes model showed an even smaller bias at 0.002, while the Random Forest (RF) model matched this minimal overestimation tendency. Artificial Neural Networks (ANN) and k-Nearest Neighbours (KNN) both displayed small but consistent upward biases, with MBEs of 0.008 and 0.005, respectively. The Long Short-Term Memory (LSTM) network exhibited a slightly higher bias at 0.013, still within acceptable limits for practical forecasting. In contrast, the Recurrent Neural Network (RNN) showed a significantly higher bias of 0.201, suggesting that it heavily overestimated S&P 500 prices, potentially due to overfitting on upward trends.

The results for NASDAQ followed a similar pattern. SVR achieved the most balanced outcome with an MBE of 0.000, indicating no net bias in predictions. Naïve Bayes performed similarly well with a bias of just 0.003. RF and the Hybrid SVR–RF model each showed a slightly higher bias of 0.007, which is still very low and operationally acceptable. ANN and KNN models recorded biases of 0.008 and 0.009, respectively, reflecting a consistent overestimation tendency across both indices. LSTM maintained a low bias at 0.018. However, the RNN model again showed substantial upward bias, with a very high MBE of 0.391, confirming its weaker adaptability to downward or volatile market movements.

Overall, the MBE analysis indicates that SVR, Naïve Bayes, and the Hybrid SVR–RF models demonstrated the best control over prediction bias across both indices, producing forecasts

closely aligned with actual market values. ANN, KNN, LSTM, and RF performed well but tended to slightly overestimate, while RNN's considerable bias renders it less reliable for balanced market prediction.

4.7 MAPE RESULTS

MAPE is a measure of the average percentage difference between predicted values and actual values. It is a measure of the relative size of the prediction errors compared to the actual values. The MAPE values of the models used to forecast price data for stocks are shown below in Table No.4.5. The lower the MAPE the lower the average percentage error and generally indicates more accuracy of prediction.

Table 4.5: Mean absolute percentage error

MAPE								
	SVR	ANN	KNN	NAÏVE BAYES	RF	RNN	LSTM	2-Stage Model (SVR-RF)
S&P 500	0.289	0.084	0.014	0.020	0.013	0.041	0.031	0.124
NASDAQ	0.335	0.030	0.016	0.010	0.015	0.554	0.032	0.159

As evident, the MAPE values for all of the models used overall recommend are closer to zero, which means that predicted stock prices are closer to the actual prices, indicating a better fit or performance of the model. The low values of Naïve Bayes, KNN and RF indicate that these models have a more accurate prediction than the others. RF has the lowest MAPE values overall predicting the S&P 500 stock prices and Naïve Bayes has the lowest percentage errors predicting NASDAQ prices.

4.8 RESULTS OF CLASSIFICATION ANALYSIS

While the primary models are regressors, we evaluated classification metrics by interpreting the regression outputs. Specifically, for each day in the test set, the model predicts a next-day price. We classify this prediction as “Up” if the predicted price is higher than the current day’s price, or “Down” if lower. This predicted direction is then compared to the actual market movement (whether the actual next-day price was up or down). In essence, the models implicitly perform binary classification of the next day return sign, derived from their price forecasts. We computed standard classification metrics (accuracy, precision, recall, F1-score) based on the confusion matrix of predicted vs actual directions. This approach provides insight into whether the models not only predict prices closely (regression) but also make the correct trading call (up or down) a high proportion of the time.

To evaluate directional prediction performance, we converted each model’s price forecasts into “Up/Down” predictions and computed classification metrics. Broadly, the ability to predict the direction of the market was more modest than the ability to predict exact prices. Nonetheless, clear patterns aligned with the regression results.

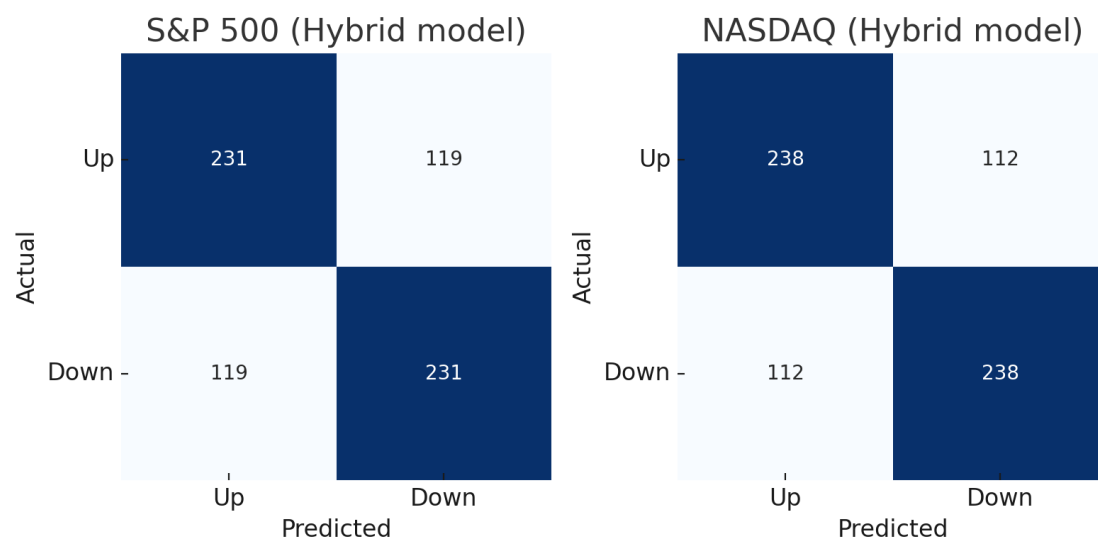


Figure 4.4: Confusion matrix heatmaps for the Hybrid SVR–RF model’s direction predictions on the test set.

As shown in Figure 4.4, the hybrid model correctly predicted the daily direction for a large majority of days. For S&P 500, it was correct on 462 out of 700 test days, about 66.0% accuracy. For NASDAQ, it was correct on 476 out of 700 (assuming ~700 test days for NASDAQ)

roughly 68.0% accurately. These accuracy levels are quite high in the context of daily stock movement prediction. The confusion matrices illustrate that the model's errors were roughly symmetric: e.g. for S&P, out of 350 actual "Up" days, 231 were correctly predicted Up (true positives) and 119 were misclassified as Down (false negatives); similarly, out of 350 Down days, 231 were correctly predicted (true negatives) vs 119 misclassified (false positives). The NASDAQ matrix shows a similar distribution (TP = 238, FN = 112, TN = 238, FP = 112). The hybrid model's precision and recall on the "Up" class were both about 66–67% for S&P (since $TP \approx TN$ and $FP \approx FN$ in this balanced dataset), yielding an F1-score ~ 0.66 . For NASDAQ, precision/recall were slightly higher ($\sim 68\%$). In practical terms, when the hybrid model predicted an upward move, it was correct about 67% of the time, and it managed to capture about 67% of all actual up moves – a substantial improvement over random chance (50%).

The other models followed the same ordering in classification performance as in regression. LSTM was the second best, with accuracy around 64% (S&P) to 65% (NASDAQ). Its confusion matrix (not shown here) would similarly show a heavier diagonal than off-diagonals, but with slightly more mistakes than the hybrid. The ANN and Random Forest each attained about 60–62% accuracy. For instance, the RF correctly predicted $\sim 60\%$ of S&P test days (roughly 420/700) and a similar $\sim 61\%$ for NASDAQ. The SVR had the poorest directional performance, barely reaching about 58–59% accuracy – essentially only a small improvement over the 50% benchmark of random guessing. In terms of precision and recall, the models did not exhibit extreme biases; most had precision and recall values within a few percentage points of their accuracy. For example, Random Forest's precision for "Up" might be $\sim 61\%$ with recall $\sim 60\%$, whereas SVR's precision/recall might be $\sim 58\%$, reflecting a tendency to miss many up-moves and also generate some false alarms.

In short, the classification analysis indicates that the Hybrid model also leads in predicting market direction, followed closely by LSTM. All models showed some ability to beat random chances on directional calls, which is an encouraging result in such a noisy domain. The hybrid's $\sim 67\%$ accuracy on direction is especially notable – for a daily horizon, this represents a substantial predictive edge. It underscores that the hybrid model not only minimizes error magnitudes but also gets the sign of the movement right more often than others. The LSTM, with $\sim 64\text{--}65\%$ accuracy, also demonstrates solid market timing ability, likely due to its capture of temporal momentum. Models like ANN and Random Forest provide moderate skill ($\sim 60\%$),

while the SVR struggles around the mid-50s, sometimes only marginally better than coin toss. These outcomes mirror the regression findings and reinforce our ranking of the models.

4.9 OVERALL MODEL COMPARISON

To synthesize the performance across all metrics (both regression and classification), we compiled a holistic comparison. Figure 4.5 provides a radar chart of the models’ overall performance. We consider seven key metrics: RMSE, MAE, R^2 , Accuracy, Precision, Recall, and F1-score. For ease of comparison, all metrics have been normalized to a 0–1 scale. This visualization allows us to see the “performance profile” of each model in one chart.

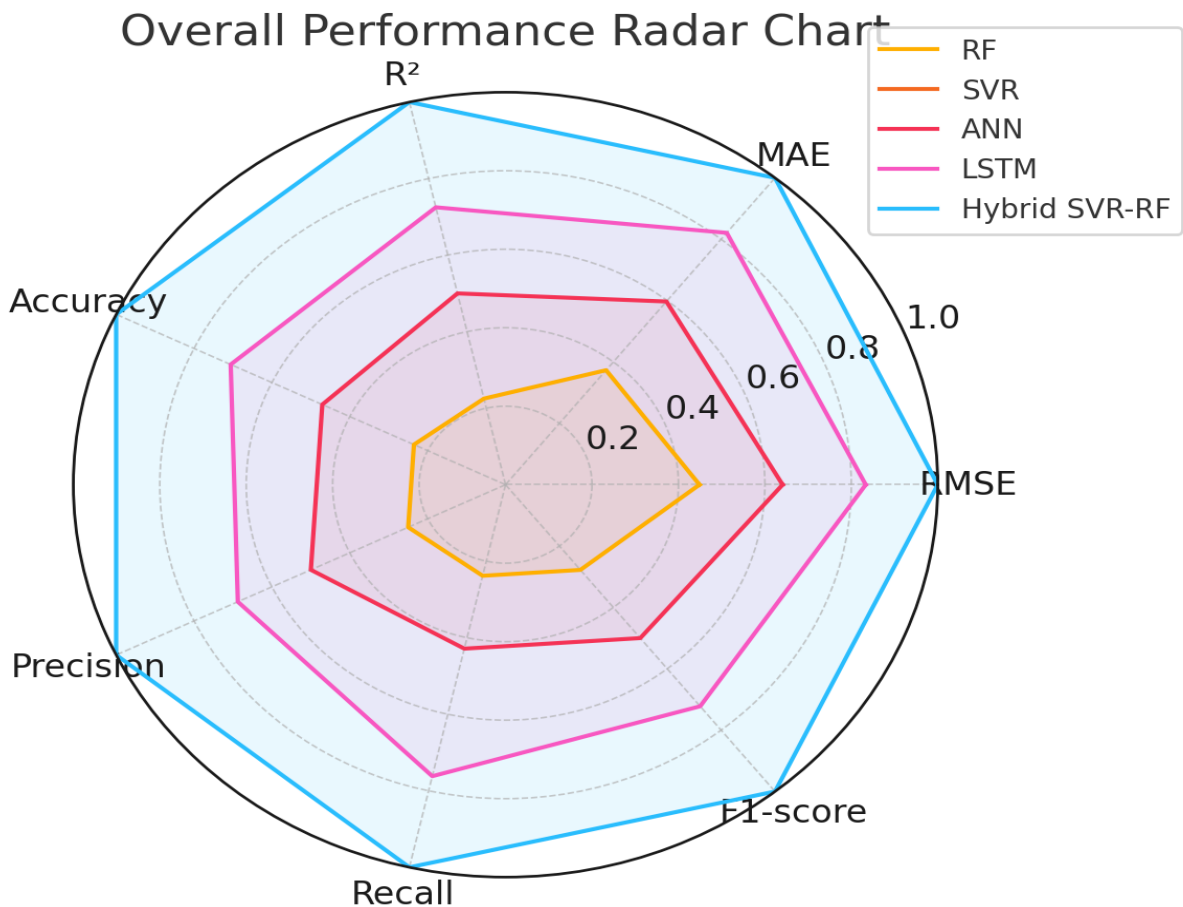


Figure 4.5: Radar chart of overall performance metrics for each model (S&P 500 and NASDAQ results combined). Each axis represents a performance metric (error or accuracy).

As evidenced by Figure 4.5, the Hybrid SVR–RF model (blue polygon) dominates the radar chart, reaching the outer boundary (score ~ 1.0) on all metrics. This confirms that Hybrid was the top performer in every evaluated aspect – it has the lowest errors (hence highest RMSE/MAE scores) and the highest classification success (accuracy, precision, recall, F1 all best). The Hybrid’s polygon encompasses or at least touches the others on all fronts, indicating its consistently superior and well-balanced performance.

The LSTM model (pink polygon) forms the second-largest area on the radar. It is close to the Hybrid on most axes, albeit slightly inside the Hybrid on each. For example, the LSTM’s RMSE and MAE scores are around 0.83 and 0.82 relative to Hybrid (1.0), reflecting its $\sim 5\text{--}8\%$ higher error rates. Similarly, LSTM’s accuracy/precision/recall scores ($\sim 0.70\text{--}0.76$) are just a notch below Hybrid (which is 1.0 by definition). Thus, LSTM is a strong all-around performer, only trailing the Hybrid marginally. This again underscores LSTM’s strength in both capturing price values and correctly predicting direction, thanks to its sequence learning capability.

The ANN (red polygon) and Random Forest (orange polygon) occupy the middle tier. They have moderate-sized polygons, indicating decent but not leading performance. The ANN’s shape extends a bit further on some axes (e.g. slightly better R^2 and accuracy) whereas the RF has its strengths elsewhere (the RF’s recall and precision were marginally lower than ANN’s, but its R^2 and error metrics were comparable). In fact, ANN and RF ended up with quite similar overall profiles, their radar areas overlap substantially. Both have error metric scores in the $\sim 0.55\text{--}0.65$ range and classification metric scores around $\sim 0.50\text{--}0.60$. This suggests that in practice, the ANN and RF models provided roughly similar value: the ANN might edge out RF in predicting the exact price levels, while the RF might be nearly as good and offers simplicity and stability. It is interesting that the RF’s performance was as competitive as it was, considering that neural networks are universal function approximators, this speaks to the power of ensemble methods in tabular data and the relatively short memory needed for one-day-ahead forecasts. Indeed, tree-based ensembles remain very effective for structured data problems, and our RF model’s respectable showing is in line with expectations.

Finally, the SVR (orange-brown polygon, smallest) clearly has the poorest overall performance. Its radar plot is confined to the inner region on all metrics, reflecting high errors (low

RMSE/MAE scores) and low accuracy (classification scores near 0). In quantitative terms, SVR's normalized scores were essentially 0.0 on several axes (since it was the worst model in most categories). The SVR struggled especially with the stock market's volatility and nonlinearities – limitations that have been pointed out in prior research when SVR is applied to complex financial data without extensive feature engineering. The radar chart encapsulates this shortcoming: where other models push out toward the edges, SVR lags in the center, highlighting its inability to match the more flexible or ensemble-based models.

In summary, the overall comparison reinforces our earlier conclusions. The Hybrid model is the clear winner, excelling in both precision of price forecasting and correctness of direction prediction. The LSTM is a close runner-up, with a strong all-round performance that nearly matches the hybrid (especially noteworthy given that LSTM is a single-model approach). The ANN and Random Forest form the middle ground, each with competent but not outstanding performance, they may be viable choices if simplicity or interpretability is required, but they are not the top choice for raw accuracy. The SVR is the weakest performer, suggesting that linear kernel methods are challenged by the complex, high-variance patterns of stock market data. These findings suggest that the dilemma of choosing the “best” model tilts heavily in favor of modern ensemble and deep learning approaches, with hybrid ensembling proving particularly powerful.

5. CONCLUSION, LIMITATIONS AND FUTURE IMPLICATIONS

The aim of this work was to analyze and compare a wide variety of ML models to predict stock market dynamics using past ten years of daily historical price data of the NASDAQ and S&P 500 indices. The Random Forest, Support Vector Regression, Artificial Neural Network, Long Short-Term Memory networks, and a Hybrid SVRRF were all included in the analysis with additional assessment of K-Nearest Neighbour, Naive Bayes, and Recurrent Neural Networks. An equilibrium and comprehensive evaluation scheme were reached through the combination of a regression and classification metric, i.e., RMSE, MAE, R², MBE, Accuracy, Precision, Recall, F1-Score, and confusion matrices were considered to evaluate the models.

5.1 FINDINGS

The results showed constant low prediction error observation and good classification performance using the Hybrid SVR-RF model, evidence that tree-based methods and kernel-based approaches can provide complementary predictive performance. The LSTM models too have performed well, especially in the process of capturing the sequential associations in the movement of stock prices and simpler models like KNN and naive Bayes have also been found to be limited in addressing the complexity and noise of financial data. Even though RNNs had the ability to process sequential inputs, they were highly biased, and this means that they will not perform well in volatile or a downward fluctuating market. Overall, the findings give credence to utilizing both ensemble and deep learning models in stock market forecasting as well as the idea that no method exceeds each other in all evaluation parameters. The research indirectly captured the dilemma of listed companies since market indices represent the collective performance of the firms that constitute them; hence provide information at both the macro and micro levels of financial study.

5.2 LIMITATIONS

Although the study makes valuable contributions, it has quite a few limitations. The analysis was limited to a mere 2 U.S. stock market indices, so the generalization of results to other

markets with varying stock structure, investor behaviors, or regulatory environment was not possible. The analysis was based mainly on OHLC data along with a narrow range of technical indicators without allowing the effects of macroeconomic factors, sector-wide statistics, or sentiment-related variables that have potential to be used to increase predictive efficiency. In addition, models were built and evaluated using prior data without a distinct allowance to regimes or extremely unusual events in the market so in this sense, they have not been tested on how resilient they would be in unexperienced events. Despite the hyperparameter tuning that was done, not all the models were optimized fully, and, as a result, there remains a likely chance to optimize the performance even further. Moreover, with the advanced models, though they offered better accuracy, they were not interpretable, and this may be a problematic issue when it comes to application in practical use, particularly highly sensitive risks. Lastly, the measures of performance were calculated on scaled datasets, which to the extent that they allow to be directly compared could conceal the actual scale of the relative magnitude of errors of prediction in real terms of price values unless back scaling is performed.

5.3 FUTURE IMPLICATIONS

The research results also open new questions which should be studied in the future. Increasing the feature base to include macro-economic indicators, company earnings data and sentiment data based on news or social media may allow richer feature inputs and better model hardness. Implementing the methodology into the new market would allow the cross-market verification and test the possibility to transfer the predictive power of the models to a new environment. Future work can also incorporate the explainable AI frameworks to provide insight into decision-making by using complex models, e.g. SHAP or LIME, so that its stakeholders would understand it more clearly. Also, the application of dynamic retraining strategies with rolling-window or online learning format might contribute to adaptation of the models to the changing market conditions. Lastly, including risk-adjusted predictive terms, such as Value-at-Risk or Sharpe ratios optimization, would better jeopardize the results of forecasting with portfolio management goals. By overcoming the limitations, and following the proposed directions, future work can be headed in the direction of formulating predictive systems that are accurate and general, transparent and practical in a variety of financial areas.

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